

Lecture 5

2nd Hybrid Class: Supervisory control

Introduction

- A control system involves four components:
 - 1- Plant**, the physical process that is to be controlled;
 - 2- The environment**, in which the plant operates;
 - 3- Sensors**, that measure some variables of the plant and the environment;
 - 4- Controller** that determines the mode transition structure and selects the time-based inputs to the plant.

Controller Levels

- The controller has two levels:
 - 1- **Supervisory control** that determines the mode transition structure,
 - 2- **Low-level control** that determines the time-based inputs to the plant.
- Where, Intuitively, the supervisory controller determines which of several strategies should be followed, and the low-level controller implements the selected strategy.

Example 4.8: Consider an automated guided vehicle (AGV) that moves along a closed track painted on a warehouse or factory floor. We will design a controller so that the vehicle closely follows the track.

The vehicle has two degrees of freedom. At any time t , it can move forward along its body axis with speed $u(t)$ with the restriction that $0 \leq u(t) \leq 10$ mph (miles per hour). It can also rotate about its center of gravity with an angular speed $\omega(t)$ restricted to $-\pi \leq \omega(t) \leq \pi$ radians/second. We ignore the inertia of the vehicle, so we assume that we can instantaneously change the velocity or angular speed.

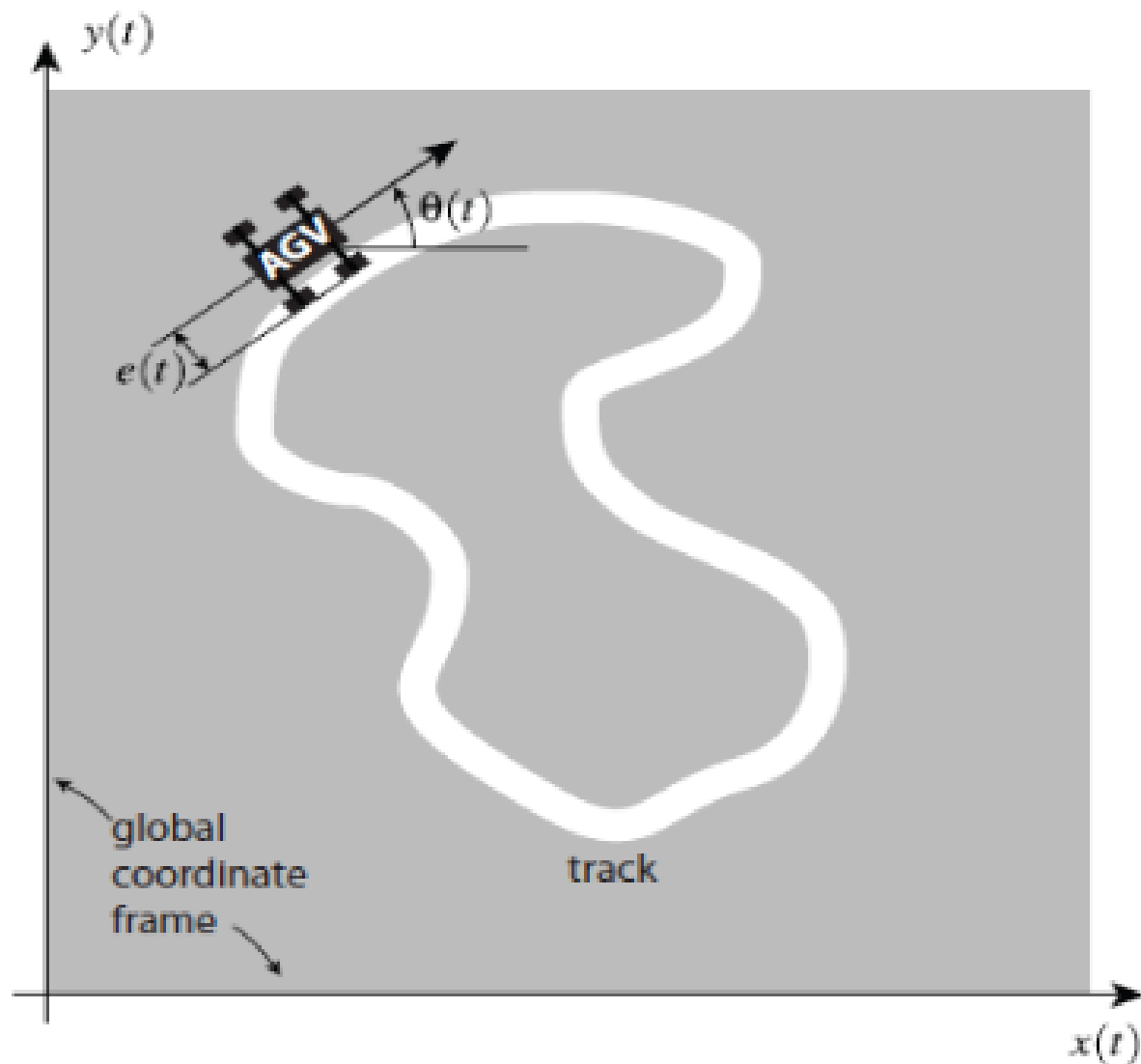


Figure 4.12: Illustration of the automated guided vehicle of Example 4.8. The vehicle is following a curved painted track, and has deviated from the track by a distance $e(t)$. The coordinates of the vehicle at time t with respect to the global coordinate frame are $(x(t), y(t), \theta(t))$.

1- Plant Description

- 1- At any time t , it can move forward along its body axis with speed $u(t)$ with the restriction that

$$0 \leq u(t) \leq 10 \text{ mph (miles per hour).}$$

- 2- It can also rotate about its center of gravity with an angular speed $\omega(t)$,

$$\text{restricted to } -\pi \leq \omega(t) \leq \pi \text{ radians/second.}$$

- 3- Let $X(t), Y(t) \in \mathbb{R}^2$, be the position of the vehicle relative to some fixed coordinate frame.

- 4- Let the angle of the vehicle at any time t is:

$$\Theta(t) \in [-\pi, \pi]$$

Cont.,

5- In terms of this coordinate frame, the motion of the vehicle is given by a system of three differential equations,

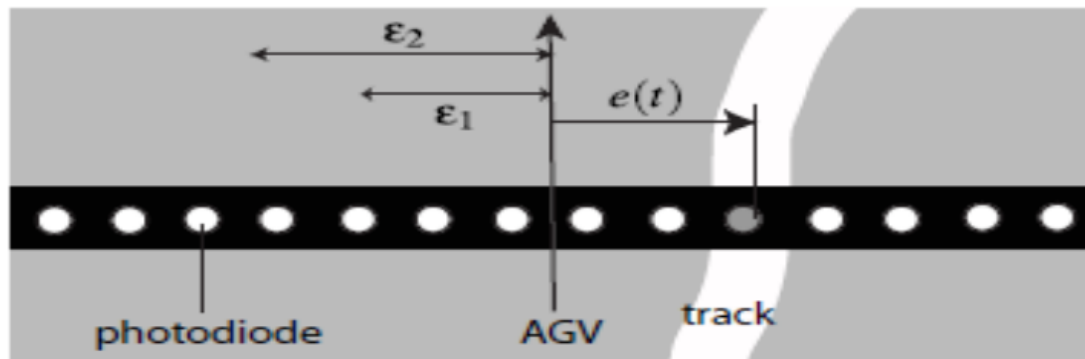
$$\begin{aligned}\dot{x}(t) &= u(t) \cos \theta(t), \\ \dot{y}(t) &= u(t) \sin \theta(t), \\ \dot{\theta}(t) &= \omega(t).\end{aligned}\tag{4.6}$$

2,3- The Environment & Sensors

- The environment is the closed painted track.
- It could be described by an equation.
- We will describe it indirectly below by means of a sensor that determines how far the vehicle is to Left or Right of the track.
- We can build such a sensor using photodiodes.
- Let's suppose the track is painted with a light-reflecting color, whereas the floor is relatively dark.
- Underneath the AGV we place an array of photodiodes as shown in the following Figure.

Cont.,

- The array is perpendicular to the AGV body axis.
- As the AGV passes over the track, the diode directly above the track generates more current than the other diodes. By comparing the magnitudes of the currents through the different diodes, the sensor estimates the displacement $e(t)$ of the center of the array (hence, the center of the AGV).



Cont.,

- We adopt the convention that $e(t) < 0$ means that the AGV is to the right of the track (θ is -ve), and $e(t) > 0$ means it is to the left (θ is +ve).
- We model the sensor output as a function f of the AGV's position,

$$\forall t, \quad e(t) = f(x(t), y(t)).$$

4-a: Low-Level Controller

- 1- The vehicle always moves at its maximum speed of 10 mph.
- 2- If the vehicle strays too far to the left of the track, the controller steers it towards the right;
- 3- If it strays too far to the right of the track, the controller steers it towards the left.
- 4- If the vehicle is close to the track, the controller maintains the vehicle in a straight direction.

Cont.,

- Thus the controller guides the vehicle in four modes: { left, right, straight, and stop}.

straight

$$\dot{x}(t) = 10 \cos \theta(t)$$

$$\dot{y}(t) = 10 \sin \theta(t)$$

$$\dot{\theta}(t) = 0$$

left

$$\dot{x}(t) = 10 \cos \theta(t)$$

$$\dot{y}(t) = 10 \sin \theta(t)$$

$$\dot{\theta}(t) = \pi$$

Cont.,

right

$$\dot{x}(t) = 10 \cos \theta(t)$$

$$\dot{y}(t) = 10 \sin \theta(t)$$

$$\dot{\theta}(t) = -\pi$$

stop

$$\dot{x}(t) = 0$$

$$\dot{y}(t) = 0$$

$$\dot{\theta}(t) = 0$$

4-b: Supervisory control

- We design the supervisory control governing transitions between modes in such a way that the vehicle closely follows the track, using a sensor that determines how far the vehicle is to the left or right of the track.
- We select two thresholds, $0 < \epsilon_1 < \epsilon_2$, as shown in Figure 4.14.
- If the magnitude of the displacement is small, $e(t) < \epsilon_1$, we consider that the AGV is close enough to the track, and the AGV can move straight ahead, in straight mode.

Cont.,

- If $e(t) > \epsilon_2$ ($e(t)$ is large and positive), the AGV has strayed too far to the left and must be steered to the right, by switching to right mode.
- If $e(t) < -\epsilon_2$ ($e(t)$ is large and negative), the AGV has strayed too far to the right and must be steered to the left, by switching to left mode.

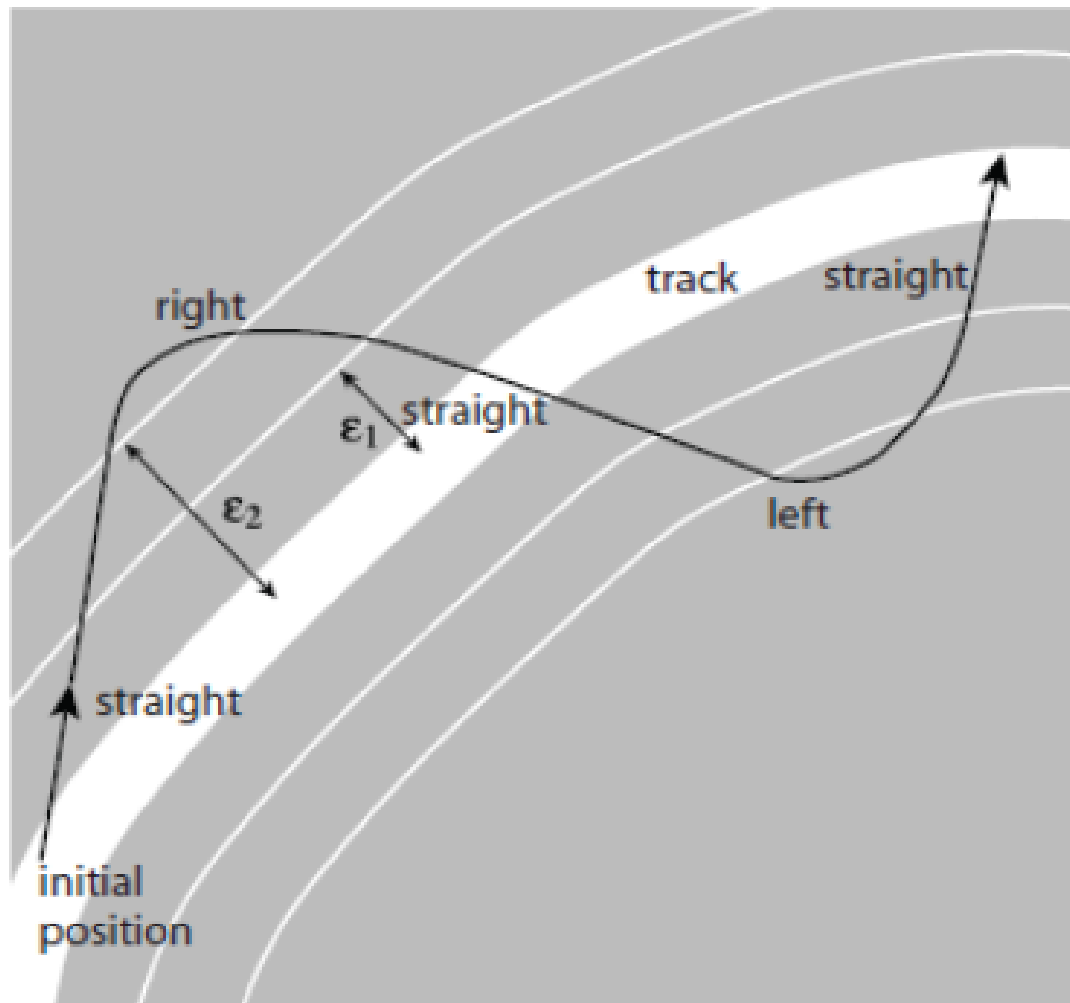
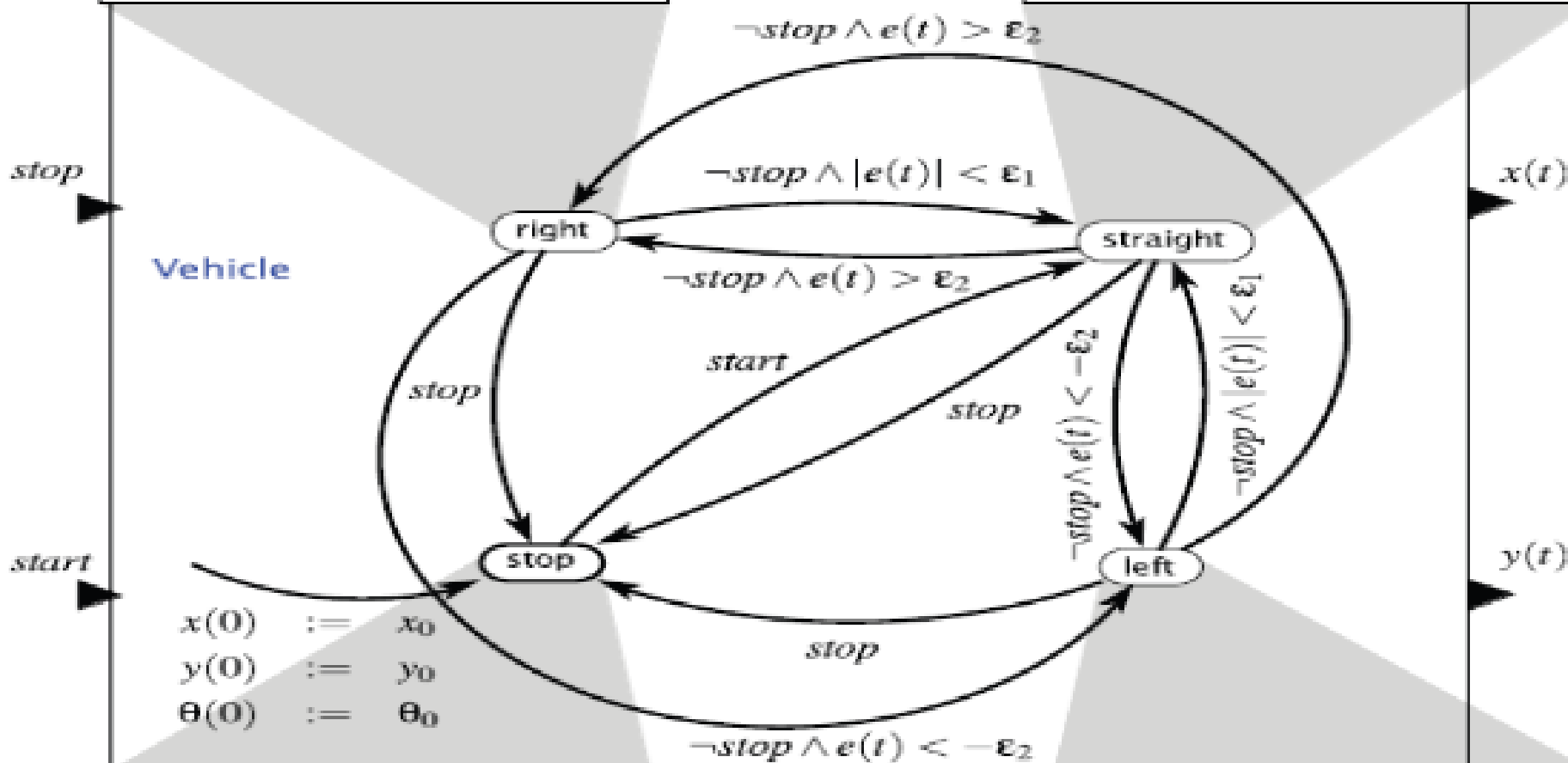


Figure 4.15: A trajectory of the AGV, annotated with modes.

$$\begin{array}{lcl} \dot{x}(t) & = & 10 \cos \theta(t) \\ \dot{y}(t) & = & 10 \sin \theta(t) \\ \dot{\theta}(t) & = & -\pi \\ e(t) & = & f(x(t), y(t)) \end{array}$$

$$\begin{array}{rcl} \dot{x}(t) & = & 10 \cos \Theta(t) \\ \dot{y}(t) & = & 10 \sin \Theta(t) \\ \dot{\Theta}(t) & = & 0 \\ e(t) & = & f(x(t), y(t)) \end{array}$$



$$\begin{array}{rcl} \dot{x}(t) & = & 0 \\ \dot{y}(t) & = & 0 \\ \dot{\Theta}(t) & = & 0 \\ e(t) & = & f(x(t), y(t)) \end{array}$$

$$\begin{array}{lcl} \dot{x}(t) & = & 10 \cos \Theta(t) \\ \dot{y}(t) & = & 10 \sin \Theta(t) \\ \dot{\Theta}(t) & = & \pi \\ e(t) & = & f(x(t), y(t)) \end{array}$$

Project # 1

- Implement the AGV,s practically.